

Reverse-check review package — Triplets_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	Triplets/model/Triplets_gen.fr
original model name	Triplets_gen (hidden from the agent)
paper	Triplets/text/0909.2666.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

`LTripKin (:=)`

```
DC[trip1bar[k], mu] DC[trip1[k], mu] - Mtrip1^2 trip1bar[k] trip1[k]
```

`LD11 (:=)`

```
2 (Eps[k, i, j] trip1bar[k] LQQR[n, m] ProjP[s, r] dqbar[s, n, i].CC[uq][r, m, j] + Eps[k,
↪ i, j] trip1bar[k] LUDL[n, m] ProjM[s, r] dqbar[s, n, i].CC[uq][r, m, j])
```

`LD1 (:=)`

```
LD11 + HC[LD11]
```

`LPot (:=)`

```
ExpandIndices[LHS1 Phibar[ii] Phi[ii] trip1bar[k] trip1[k] + LSS11 trip1bar[k1] trip1[k1]
↪ trip1bar[k2] trip1[k2], FlavorExpand -> {SU2W, SU2D}]
```

`LTrip (:=)`

```
LTripKin + LD1 + LPot
```

Blank-slate reconstruction

Reconstructed Physics From `sanitized.fr`

Lagrangian

Let the new scalar field be

$$S_k \equiv \text{trip1}_k,$$

with color index $k = 1, 2, 3$. Its conjugate is $S_k^\dagger \equiv \text{trip1bar}_k$. The Standard Model Higgs doublet is Φ_i .

The scalar is an $SU(3)_c$ triplet, $SU(2)_L$ singlet, with

$$Y_S = Q_S = -\frac{1}{3}.$$

The covariant derivative acting on S is

$$D_\mu S_k = \partial_\mu S_k + ig_s G_\mu^a (T^a)_k^\ell S_\ell + ig' Y_S B_\mu S_k,$$

with no $SU(2)_L$ term because S is an electroweak singlet.

LTripKin

$$\mathcal{L}_{\text{LTripKin}} = (D_\mu S)_k^\dagger (D^\mu S)_k - M_{\text{trip1}}^2 S_k^\dagger S_k.$$

LD11

The file defines

$$\lambda_{nm}^{QQ} \equiv \text{LQQR}[n, m], \quad \lambda_{nm}^{UD} \equiv \text{LUDL}[n, m].$$

Using $P_R = (1 + \gamma^5)/2$, $P_L = (1 - \gamma^5)/2$, and $u^c = C\bar{u}^T$, the interaction term is

$$\mathcal{L}_{\text{LD11}} = 2 \epsilon_{kij} S_k^\dagger \lambda_{nm}^{QQ} \bar{d}_{ni} P_R u_{mj}^c.$$

Because $P_R u^c = (u_L)^c$, this is equivalently a left-chiral QQ -type diquark coupling:

$$2 \epsilon_{kij} S_k^\dagger \lambda_{nm}^{QQ} \overline{d_{L,ni}} (u_{L,mj})^c.$$

The second term is

$$\mathcal{L}_{\text{LD11}} \supset 2 \epsilon_{kij} S_k^\dagger \lambda_{nm}^{UD} \bar{d}_{ni} P_L u_{mj}^c.$$

Because $P_L u^c = (u_R)^c$, this is equivalently a right-chiral $u_R d_R$ -type diquark coupling:

$$2 \epsilon_{kij} S_k^\dagger \lambda_{nm}^{UD} \overline{d_{R,ni}} (u_{R,mj})^c.$$

Here i, j, k are color indices contracted with the antisymmetric $SU(3)_c$ tensor ϵ_{kij} , and n, m are generation indices.

LD1

$$\mathcal{L}_{\text{LD1}} = \mathcal{L}_{\text{LD11}} + \mathcal{L}_{\text{LD11}}^\dagger.$$

Explicitly, the hermitian conjugate contains the corresponding S_k couplings to conjugated quark bilinears.

LPot

$$\mathcal{L}_{\text{LPot}} = \lambda_{HS} (\Phi_i^\dagger \Phi_i) (S_k^\dagger S_k), \quad \lambda_{HS} \equiv \text{LHS1}.$$

$$\mathcal{L}_{\text{LPot}} \supset \lambda_{SS} (S_{k_1}^\dagger S_{k_1}) (S_{k_2}^\dagger S_{k_2}), \quad \lambda_{SS} \equiv \text{LSS11}.$$

LTrip

$$\mathcal{L}_{\text{LTrip}} = \mathcal{L}_{\text{LTripKin}} + \mathcal{L}_{\text{LD1}} + \mathcal{L}_{\text{LPot}}.$$

Field Table

.fr symbol	Particle	Spin	$SU(3)_c$	$SU(2)_L$	Y	Q	Self- conjugate	Mass
trip1	complex scalar S	0	triplet	singlet	$-1/3$	$-1/3$	no	Mtrip1 = 500.

The antiparticle is **trip1~**, represented in the Lagrangian as **trip1bar**.

Parameters

Symbol	Type	Value	Multiplies	Physical meaning
LQRR[n,m]	external real matrix	nonzero entries: $LQRR_{12} = 0.1,$ $LQRR_{21} = -0.1,$ $LQRR_{13} = 0.1,$ $LQRR_{31} = -0.1,$ $LQRR_{23} = 0.1,$ $LQRR_{32} = -0.1,$ diagonal 0	real part of LQQR[n,m] in $S^\dagger \bar{d} P_R u^c$	real part of left-chiral QQ -type diquark coupling
LQRI[n,m]	external real matrix	all entries 0	imaginary part of LQQR[n,m]	imaginary part / CP phase of left-chiral diquark coupling
LUDLR[n,m]	external real matrix	$LUDLR_{11} =$ $LUDLR_{22} =$ $LUDLR_{33} = 0.1,$ off-diagonal 0	real part of LUDL[n,m] in $S^\dagger \bar{d} P_L u^c$	real part of right-chiral $u_R d_R$ -type diquark coupling
LUDLI[n,m]	external real matrix	all entries 0	imaginary part of LUDL[n,m]	imaginary part / CP phase of right-chiral diquark coupling
LHS1	external real scalar	1.0	$(\Phi^\dagger \Phi)(S^\dagger S)$	Higgs-portal quartic coupling
LSS11	external real scalar	1.0	$(S^\dagger S)^2$	scalar self-quartic coupling

The internal complex couplings are

$$\text{LQQR}_{nm} = \text{LQRR}_{nm} + i \text{LQRI}_{nm},$$

$$\text{LUDL}_{nm} = \text{LUDLR}_{nm} + i \text{LUDLI}_{nm}.$$

Physics Summary

This is a Standard Model extension by one complex color-triplet, electroweak-singlet scalar with electric charge $-1/3$. It couples as a diquark to up- and down-type quarks through both left-chiral QQ -type and right-chiral $u_R d_R$ -type bilinears, and it also has Higgs-portal and scalar self-interactions.

The model mediates baryon-number-sensitive quark-quark interactions such as resonant or virtual scalar exchange in ud -type partonic channels, including dijet production and flavor-changing quark processes depending on the generation structure of LQQR and LUDL.

Paper cross-check

Comparison of reconstruction.md Against the Paper Model

Paper Model Location

The model is defined in **Section 2, “Model And Constraints”**. The electroweak field assignments are summarized in **Table 1**, and the post-EWSB diquark-quark interaction Lagrangian is given in **Eq. (2.1)**:

$$L = 2\sqrt{2} [\bar{K}_{ab}^i D_i \bar{q}^a (\lambda_L P_L + \lambda_R P_R) q_b^C + \text{h.c.}] .$$

The paper states that D_i is a scalar diquark in either the **sextet** or **antitriplet** of $SU(3)_C$, with Clebsch-Gordan coefficients $\bar{K}_{ab}^i$ coupling the diquark representation to two color triplets. The color conventions are detailed in **Appendix A**, especially **Eqs. (A.6)-(A.10)**. For the antitriplet case, **Eq. (A.7)** and the following paragraph give

$$K_{abc} = \epsilon_{abc}/\sqrt{2}.$$

The paper also states in Section 2 that, in the antitriplet case, the couplings must be antisymmetric in flavor.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Possible diquark electroweak assignments: Table 1 lists $SU(2)_L = 1$, $Y = 1/3$, $ Q = 1/3$, coupling to QQ, UD ; also other possibilities $SU(2)_L = 3, Y = 1/3$, $SU(2)_L = 1, Y = 2/3$, $SU(2)_L = 1, Y = 4/3$.	Reconstruction selects one scalar S : $SU(3)_c$ triplet, $SU(2)_L$ singlet, $Y = Q = -1/3$.	disagree	The reconstruction corresponds to the conjugate of the paper’s $SU(2)$ -singlet $Y = +1/3$ diquark if one identifies $D = S^\dagger$. As written, the field table assigns the opposite hypercharge/charge to the named scalar relative to the paper’s D . The paper is also generic over antitriplet or sextet color, while the reconstruction fixes the antitriplet/conjugate-triplet case.
Diquark color representation: Section 2 says D_i transforms as either 6 or $\bar{3}$ of $SU(3)_C$; Appendix A gives both symmetric sextet and antisymmetric antitriplet Clebsch-Gordan tensors.	Reconstruction has a single complex color-triplet scalar S_k , with $S_k^\dagger$ entering the diquark coupling via ϵ_{kij} .	agree	For the antitriplet case only, this is equivalent if $D_k \equiv S_k^\dagger$. The reconstruction does not represent the sextet option discussed in the paper.
Antitriplet Clebsch-Gordan structure: Appendix A, Eq. (A.7) and following text, $K_{abc} = \epsilon_{abc}/\sqrt{2}$.	$2 \epsilon_{kij} S_k^\dagger \bar{d}_{ni} P_{R/L} u_{mj}^c$.	agree	Combining the paper’s coefficient $2\sqrt{2}$ in Eq. (2.1) with $K = \epsilon/\sqrt{2}$ gives an overall factor 2ϵ , matching the reconstruction for the antitriplet case, up to the identification $D = S^\dagger$.

paper term (eq. ref)	reconstruction term	verdict	notes
Generic quark-diquark interaction: Eq. (2.1), $2\sqrt{2}\bar{K}_{ab}^i D_i \bar{q}^a (\lambda_L P_L + \lambda_R P_R) q_b^C + \text{h.c.}$	LD11: $2\epsilon_{kij} S_k^\dagger \lambda_{nm}^{QQ} \bar{d}_{ni} P_R u_{mj}^c$ plus $2\epsilon_{kij} S_k^\dagger \lambda_{nm}^{UD} \bar{d}_{ni} P_L u_{mj}^c$, with h.c. in LD1.	agree	This matches the paper's antitriplet specialization and the QQ/UD singlet row of Table 1, provided $D = S^\dagger$. The paper keeps q and flavor sums generic, while the reconstruction spells out d, u^c components.
Chirality projectors: Eq. (2.1), $P_{L,R} = (1 \mp \gamma^5)/2$, acting on q^C .	Reconstruction interprets $\bar{d}P_R u^c$ as a left-chiral QQ -type coupling and $\bar{d}P_L u^c$ as a right-chiral UD -type coupling.	agree	Because charge conjugation flips chirality, this interpretation is physically consistent. The reconstruction's parameter names LQQR and LUDL are notation-specific and should not be read as contradicting the physical chirality.
Hermitian conjugate: Eq. (2.1) explicitly includes +h.c..	LD1 = LD11 + LD11 [†] .	agree	Direct match.
Flavor structure: Section 2 says flavor sums are suppressed and $\lambda_{L,R}$ may be independent or model-constrained; in the antitriplet case, couplings must be antisymmetric in flavor.	Reconstruction has antisymmetric nonzero LQQR off-diagonal entries, but LUDL is diagonal.	disagree	The LQQR pattern is consistent with an antisymmetric QQ antitriplet coupling. The diagonal LUDL entries may be acceptable if treated as U - D species couplings rather than identical-flavor antisymmetry, but the paper's broad statement that antitriplet couplings are antisymmetric in flavor makes this a point requiring human confirmation.
Gauge interactions of colored scalar: Section 2 says any nontrivial $SU(3)_C$ state interacts with gluons; Section 4 and Appendix A use the diquark representation generators and Casimir C_D , with $C_D = 4/3$ for antitriplet and $10/3$ for sextet.	LTripKin: $(D_\mu S)^\dagger (D^\mu S) - M^2 S^\dagger S$, with $SU(3)_c$ and hypercharge gauge terms, no $SU(2)_L$.	agree	The kinetic/mass term is not written explicitly in Eq. (2.1), but is implied by the paper's production calculation for a massive colored scalar. The reconstruction fixes the antitriplet/conjugate-triplet and singlet-hypercharge case.

paper term (eq. ref)	reconstruction term	verdict	notes
Diquark mass: production formulas use m_D , e.g. Eqs. (3.3)-(3.4).	Mass term $-M_{\text{trip1}}^2 S^\dagger S$, with $M_{\text{trip1}} = 500$.	agree	The paper treats m_D as a variable mass parameter; the reconstruction gives one implementation value.
Higgs portal quartic ($\Phi^\dagger \Phi$)($S^\dagger S$).	LPot: $\lambda_{HS}(\Phi^\dagger \Phi)(S^\dagger S)$.	extra-in-reconstruction	The paper does not define this scalar-potential term. It may exist in a UV-complete model, but it is outside the paper's stated production Lagrangian.
Scalar self-quartic ($S^\dagger S$) ² .	LPot: $\lambda_{SS}(S^\dagger S)^2$.	extra-in-reconstruction	The paper does not specify a scalar self-potential.
Sextet diquark option: Section 2 and Appendix A include the color sextet; Table 1 says each electroweak state may be $\bar{3}$ or 6.	No sextet field or symmetric Clebsch-Gordan coupling appears.	missing-in-reconstruction	The reconstruction only covers the antitriplet/conjugate-triplet implementation, not the paper's full generic color scope.
Other Table 1 electroweak possibilities: $SU(2)_L = 3, Y = 1/3$ coupling to QQ , singlet $Y = 2/3$ coupling to DD , singlet $Y = 4/3$ coupling to UU .	No $SU(2)_L$ triplet, DD , or UU couplings.	missing-in-reconstruction	The paper lists these for completeness but then uses a generic post-EWSB parameterization. The reconstruction is a specific QQ/UD , $SU(2)$ -singlet model.

Disagreements and Checks

1. Named scalar charge and representation: substantive.

The reconstruction names the fundamental scalar S as a color triplet with $Y = Q = -1/3$, while the paper's QQ/UD diquark in Table 1 has $Y = +1/3$ and color $\bar{3}$ or 6; a human should check whether the implementation intentionally defines $S^\dagger$ as the paper's produced diquark D .

2. Sextet omission: substantive.

The paper's model parameterization covers both color antitriplet and sextet diquarks, but the reconstruction only contains the antitriplet/conjugate-triplet epsilon structure; a human should check whether the implementation was intended to reproduce only the antitriplet benchmark rather than the full paper.

3. Other electroweak assignments omitted: convention/substantive depending on scope.

Table 1 includes QQ triplet, DD , and UU singlet possibilities, but the reconstruction only implements the $SU(2)_L$ -singlet QQ/UD case; a human should check whether the implementation target was the full table or one benchmark row.

4. LUDL flavor symmetry: substantive if the paper's antitriplet antisymmetry is meant to apply to all flavor matrices.

The reconstruction gives diagonal right-chiral UD couplings, while the paper says antitriplet couplings must be antisymmetric in flavor; a human should check how the implementation defines flavor indices for U - D couplings and whether antisymmetry is required for non-identical up/down species.

5. Scalar potential terms: substantive relative to the paper text, likely harmless for production.

The reconstruction includes Higgs-portal and scalar self-quartic interactions not specified in the paper;

a human should check whether these are implementation conveniences/default scalar potential terms or intended claims about the paper’s model.

Overall Assessment

The reconstruction captures the core antitriplet QQ/UD diquark interaction of Eq. (2.1) well once the paper’s diquark is identified with $S^\dagger$: the epsilon color tensor, the overall factor of 2, the hermitian conjugate, and the chirality interpretation are all consistent with the paper’s antitriplet Clebsch-Gordan normalization. The main caveats are scope and convention: the paper is written as a generic scalar diquark parameterization covering both $\bar{3}$ and 6 color representations and multiple electroweak assignments, while the reconstruction is a specific singlet antitriplet/conjugate-triplet implementation with the named scalar carrying the opposite charge from the paper’s D . The added scalar potential terms are not part of the paper’s defined production Lagrangian, and the right-chiral UD flavor structure deserves a targeted check against the implementation’s flavor-index convention.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field’s representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 9 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model’s identity.

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: [] approve [] approve with corrections [] reject
- Notes: